

One Benchmark, Three Numbers: What Conversational Memory Scores Actually Measure

Graphon vs mem0 on LOCOMO and LongMemEval-S, in one open harness
Graphon AI | 2026-07-16 | fully reproducible: <https://github.com/arbaazkhan2graphon/graphon-memory-bench>

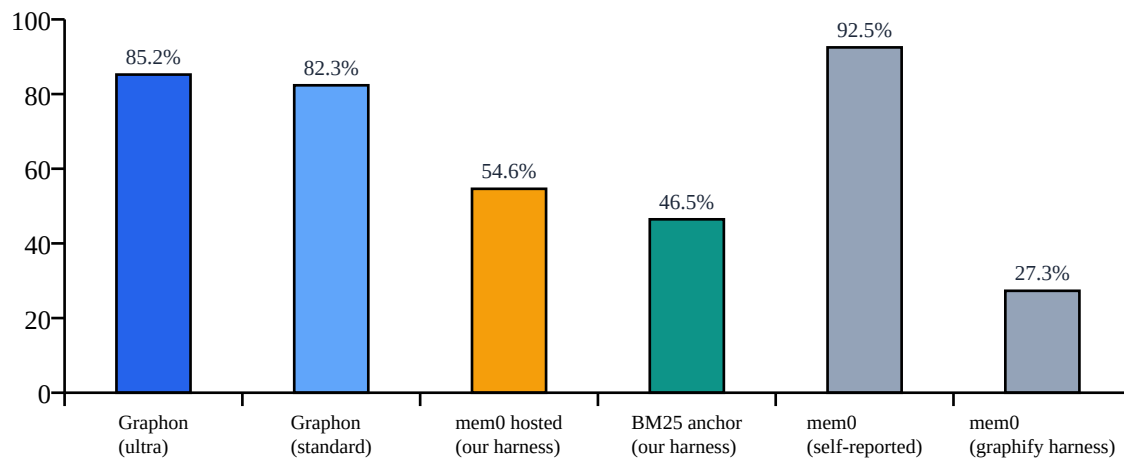
mem0 reports **92.5%** on the LOCOMO benchmark. The graphify benchmark page reports mem0 at **27.3%** on the same dataset. When we ran mem0's hosted platform ourselves, we measured **54.6%**. All three numbers are real. They differ because a benchmark score is a property of the *evaluation harness* — the judge, the retrieval depth, the prompts — at least as much as of the memory system. This report publishes our full harness, applies the same strict rules to every system including our own, and links the code so you can reproduce every number below.

Results

QA accuracy is key-fact coverage: gold answers are decomposed into atomic facts; a gpt-4o judge must cite a verbatim quote from the candidate answer for every fact it credits (half credit for partials). One shared model answers and judges for every system. LOCOMO n=300, LongMemEval-S n=50, both stratified with a fixed seed.

System	LOCOMO	LongMemEval-S
Graphon — ultra mode (agentic answer harness)	85.2%	79.5%
Graphon — standard mode	82.3%	62.5%
mem0 hosted platform (run by us, same harness)*	54.6%	—
BM25 anchor (in-harness sanity baseline)	46.5%	54.5%

* mem0 on LongMemEval-S is pending: it requires ~2,250 session ingests through their platform API. The harness supports it (--backends mem0); anyone can run it from the repository.



LOCOMO QA accuracy. Colored bars were measured in our open harness under identical grading; gray bars are external numbers for the same mem0 system under two other harnesses — the spread is the point.

Why doesn't mem0 show 92.5% here?

The same mem0 product carries three public LOCOMO numbers. Nothing about the memory system changes between them — only the way answers are graded and how much retrieved context the answering model gets:

	mem0's harness	This harness	graphify's harness
Reported score	92.5%	54.6%	27.3%
Who ran it	mem0	Graphon (code public)	graphify
Judge	Binary CORRECT/WRONG, instructed to be lenient ('same topic' can pass)	Per-fact verdicts; verbatim quote required for credit; partial = half credit	Key-fact coverage (Kimi K2.6 judge)
Retrieval depth	Up to top-200 memories	Top-10	Top-10
Answer prompts	Tuned per benchmark	One neutral prompt, identical for all systems	One shared reader

Judge leniency is the biggest lever. The 'LLM-as-a-judge' score mem0 headlines is not part of the original LOCOMO benchmark (which used lexical F1/ROUGE); it was introduced by mem0's own paper. Their judge prompt instructs the model to be generous and accept answers on the same topic as the gold answer. An independent audit (Penfield Labs, April 2026) fed that judge deliberately wrong but topically adjacent answers for all 1,540 LOCOMO questions: it accepted ~62% of them as correct. Our judge cannot do that — it must quote the exact words in the answer that establish each gold fact, and every verdict is stored in the run artifacts for inspection.

Retrieval depth changes the task. mem0's headline number gives the answering model up to 200 retrieved memories per question; at top-10 — the depth the field's comparative tables use — the task is much harder. We ran every system, including our own, at top-10.

Prompt engineering inflates self-reported numbers. A third-party reproduction (maximem, 2026) ran mem0's hosted product on LongMemEval through a standardized harness and measured 73.8% against mem0's published 93.4% on the same data — attributing the gap to benchmark-specific answer and judge prompts, not the memory layer. Our harness uses one neutral reader prompt for every system.

The same strict rules cut against us too. Under lenient same-topic grading our numbers would also be far higher. We publish the strict numbers because they are the ones that predict real product behavior: a memory system that returns the right topic but the wrong fact is wrong.

The BM25 anchor keeps everyone honest. A plain BM25 baseline runs inside the harness over the identical corpus. If you re-run this benchmark with a different judge model and get different absolute numbers, the anchor moves with them — compare system-minus-anchor deltas across harnesses.

How we ran mem0 (fairness checklist)

1. Their production hosted platform via their official SDK — not a re-implementation. One mem0 user per conversation.

2. Identical source conversations, session dates preserved (passed as timestamps and inline), mirroring mem0's own LOCOMO evaluation format.
3. Ingestion allowed to fully settle before any question was asked: we polled their memory counts until stable (their extraction runs asynchronously and lags add-calls by minutes) — 2,439 extracted memories across the 10 conversations.
4. Top-10 memory search per question; the same gpt-4o reader answered from mem0's memories and from Graphon's sources with the same prompt; the same judge graded both.
5. Retrieval-recall metrics are not reported for mem0: their memories are extracted facts rather than verbatim transcript, so our n-gram turn recall would under-credit them. QA accuracy is the comparable column.

Graphon numbers: 'standard' is the default query mode; 'ultra' is Graphon's agentic answer harness, which reasons over the memory graph before answering. Both are graded identically to mem0.

Reproduce every number

The complete harness — dataset downloaders (pinned SHAs), ingestion, backends, judge prompts, and per-question artifacts — is public:

<https://github.com/arbaazkhan2graphon/graphon-memory-bench>

```
git clone https://github.com/arbaazkhan2graphon/graphon-memory-bench.git
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt
cp .env.example .env # add GRAPHON_API_KEY, OPENAI_API_KEY (+ MEM0_API_KEY)
.venv/bin/python download_data.py
.venv/bin/python run_eval.py --benchmark locomo --backends graphon,bm25,mem0
.venv/bin/python run_eval.py --benchmark locomo --backends graphon --graphon-modes direct
--reasoning-effort ultra --no-resume
.venv/bin/python run_eval.py --benchmark longmemeval --backends graphon,bm25
```

Every run writes one JSONL row per question with the answer, the per-fact verdicts, and the verbatim quotes the judge relied on — the grading is auditable line by line, and the spend ledger caps OpenAI cost per run. Sampling is seeded, so you will evaluate the exact question set behind this report.

Sources: mem0 92.5% — mem0.ai research page and mem0ai/memory-benchmarks (top_200, April 2026). mem0 27.3% — graphify benchmark page (2026-07-05). Judge-leniency audit — Penfield Labs (dev.to, April 2026). Claimed-vs-observed reproduction — maximem.ai, 'The state of AI memory in 2026'. LOCOMO — Maharana et al., ACL 2024 (snap-research/locomo). LongMemEval-S — xiaowu0162/longmemeval-cleaned. Graphon LOCOMO standard n=299 of 300 (one question lost to a transient API error in the archived standard run; the ultra run covers all 300).